

MARATHON, FL, USA

WMO: 722016

Lat: 24.726N

Lon: 81.052W

Elev: 2

StdP: 101.30

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 12896

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.0	13.7	3.5	4.9	13.8	6.2	5.9	15.2	9.7	19.9	8.9	18.5	4.7	20	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.5	33.4	26.6	32.9	26.5	32.5	26.4	27.9	31.8	27.4	31.4	27.1	31.1	3.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.6	30.1	26.3	21.7	29.7	26.1	21.4	29.7	88.2	32.0	86.6	31.5	85.0	31.2	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.6	7.8	7.1	DB	9.1	34.7	2.1	1.2	7.5	35.6	6.3	36.3	5.1	36.9	3.5	37.8	(4)
			WB	5.5	28.2	2.2	0.5	3.9	28.6	2.7	28.9	1.5	29.2	-0.1	29.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.1	21.1	22.5	23.8	25.9	27.7	29.2	29.7	29.8	29.1	27.4	24.3	22.8
	DBStd	3.80	3.78	3.29	2.75	2.00	1.53	1.37	1.19	1.24	1.23	1.89	2.53	3.44
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	41	20	8	3	0	0	0	0	0	0	0	1	9
	CDD10.0	5883	345	351	427	476	547	576	610	613	572	539	430	396
	CDD18.3	2881	106	126	171	226	289	326	352	355	322	281	181	146
	CDH23.3	29198	424	600	1013	1843	3064	4106	4600	4705	4014	2928	1144	756
	CDH26.7	10652	31	68	153	419	966	1786	2162	2273	1693	871	158	72
Wind	WSAvg	3.6	3.9	3.7	3.9	3.9	3.6	3.0	2.8	2.9	3.1	4.0	4.2	3.9
Precipitation	PrecAvg	1036	50	43	39	50	73	117	93	150	179	150	45	46
	PrecMax	1406	209	154	90	156	231	373	212	491	385	437	169	137
	PrecMin	706	0	1	8	2	12	6	21	27	92	39	0	1
	PrecStd	187	51	40	25	38	61	85	47	85	62	105	46	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.6	29.6	30.2	31.8	32.8	34.1	34.0	34.0	33.4	32.2	30.6	29.4
		MCWB	23.5	23.9	24.7	25.4	25.8	26.5	26.7	26.9	26.8	26.4	25.3	24.6
	2%	DB	27.5	28.3	29.1	30.6	32.0	33.0	33.3	33.3	32.7	31.4	29.2	28.2
		MCWB	22.9	23.5	24.0	24.5	25.5	26.3	26.5	26.8	26.7	25.9	24.4	23.8
	5%	DB	26.7	27.5	28.3	29.8	31.2	32.4	32.8	32.9	32.3	30.9	28.3	27.5
		MCWB	22.4	22.9	23.4	23.9	25.0	26.2	26.4	26.7	26.5	25.6	23.8	23.4
	10%	DB	25.8	26.6	27.5	28.9	30.4	32.0	32.3	32.4	31.6	30.1	27.5	26.7
		MCWB	21.9	22.3	22.8	23.4	24.5	26.1	26.3	26.5	26.2	25.1	23.4	22.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.6	24.9	25.7	26.6	27.1	28.0	28.3	28.3	28.2	27.5	26.5	25.6
		MCDB	27.5	28.0	28.8	30.4	31.4	31.8	32.2	32.5	31.7	30.1	28.3	28.1
	2%	WB	23.9	24.1	24.8	25.5	26.2	27.4	27.6	27.8	27.5	26.8	25.7	24.7
		MCDB	26.5	27.4	28.0	29.2	30.6	31.4	31.7	32.0	31.2	29.7	28.0	27.1
	5%	WB	23.2	23.6	24.2	24.8	25.6	26.9	27.1	27.3	27.1	26.3	24.9	24.2
		MCDB	25.7	26.6	27.4	28.5	29.9	31.1	31.4	31.6	30.8	29.5	27.5	26.5
	10%	WB	22.6	23.0	23.6	24.2	25.2	26.5	26.7	26.9	26.7	25.8	24.2	23.6
		MCDB	25.1	25.8	26.8	27.9	29.3	30.8	31.1	31.2	30.4	28.9	26.8	26.0
Mean Daily Temperature Range	5% DB	MDBR	5.6	5.5	5.8	5.5	5.3	5.3	5.4	5.5	5.3	4.7	4.8	5.0
		MCDBR	5.6	5.7	6.0	5.8	5.7	5.6	5.7	5.8	5.6	5.1	4.9	4.8
		MCWBR	3.2	3.1	3.2	2.9	2.7	2.5	2.6	2.7	2.6	2.4	2.6	2.9
	5% WB	MCDBR	5.7	5.5	5.9	5.5	5.3	5.3	5.5	5.7	5.3	4.8	4.9	4.6
		MCWBR	3.7	3.4	3.5	3.2	2.7	2.8	2.8	3.0	2.9	2.6	3.0	3.1
Clear-Sky Solar Irradiance	taub	0.363	0.378	0.384	0.402	0.425	0.445	0.452	0.435	0.424	0.409	0.379	0.371	
	taud	2.548	2.508	2.477	2.405	2.391	2.408	2.386	2.458	2.456	2.484	2.516	2.546	
	Ebn at Noon	896	906	915	900	871	848	842	859	864	862	871	873	
	Edh at Noon	92	102	110	121	122	119	122	113	111	103	94	89	
All-Sky Solar Radiation	RadAvg	3.90	4.81	5.91	6.67	6.72	6.23	6.36	6.05	5.36	4.82	4.16	3.67	
	RadStd	0.29	0.31	0.24	0.29	0.59	0.48	0.30	0.24	0.25	0.34	0.20	0.31	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.50	+1.07	N/A	+0.28	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.38	+0.87	N/A	N/A	N/A	N/A	N/A	-13	+116	+104

Nomenclature: See separate page